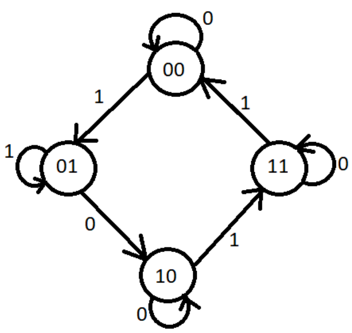
State Diagram to Circuit

1. Given the state diagram as follows, **determine** the sequential circuit using SR flipflop



Counters

2. **Design** a 3-bit odd up counter, that is: 1 → 3 → 5→7→1 (SR & JK)
3. **Design** a 3-bit even up counter, that is: 0 →2→4→6→0 (SR & JK)
4. **Design** the following counter using a) SR FF & b) JK FF : 1 → 2→5→7→1

For the missing state in all of the problems above, the next state will be the initial state.

5. **Implement** the following counter using T flip flop

CSE110 -> CSE111 -> CSE220 -> CSE221 -> CSE331 -> CSE221 -> CSE321 -> CSE110

Memory

1. **Draw** the diagram of the a) 4096×64 RAM showing internal details. Determine:

   A. Capacity of the RAM in Kilobytes or Megabytes
   B. Number of Address Lines
   C. Number of FF for Address Register, Buffer Registers
   D. Number of Data Input & Output Lines.

Variation in the configuration of the RAM may be as 2048X64, 1024X32, 512X64 … … the underlying principle remains the same.

2. How many address lines do we need for a 64 MB RAM with 16 bit/words?